

Competitive Transaction Admission in PCNs

Online Knapsack with Positive and Negative Items

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Outline

Payment Channel Networks

Problem and Model

Related Work

Algorithm Exp

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Blockchain: A Scalability Problem ¹

Bitcoin

~7

transactions / second

Visa

~1,700

transactions / second

⇒ as is, the blockchain cannot be used for daily payments.

¹[Croman + 2016] Croman et al., *On Scaling Decentralized Blockchains*, FC 2016.

A Solution: Payment Channel Networks (PCNs)

Say node s wants to send 1.5 bitcoin to node d .

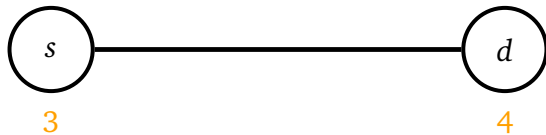
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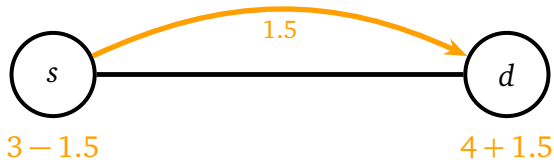
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s and d can afterwards freely exchange as long as their balance remains within the allocated bounds.

Payment Paths



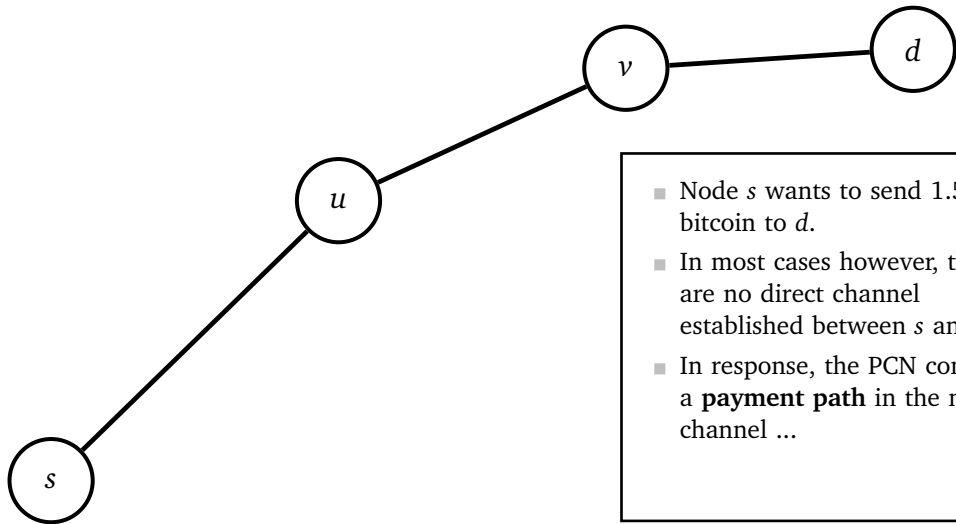
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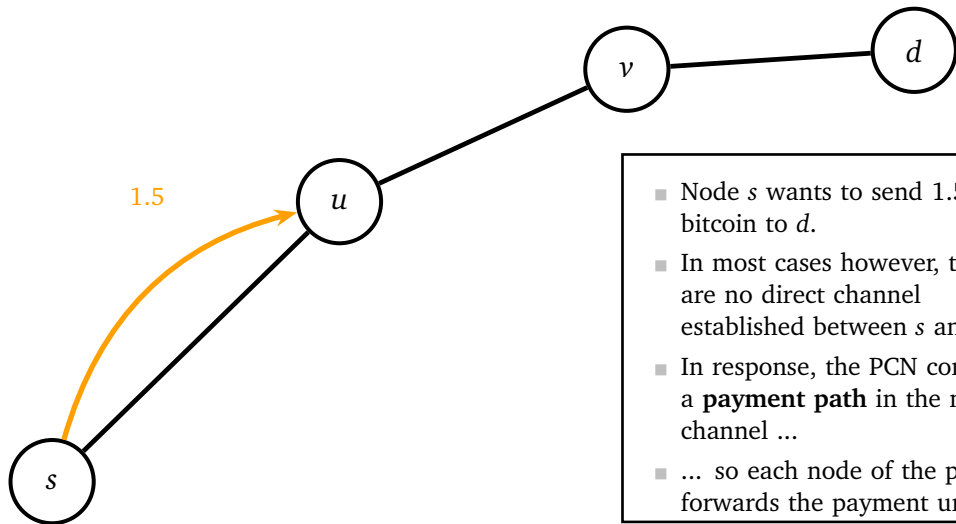
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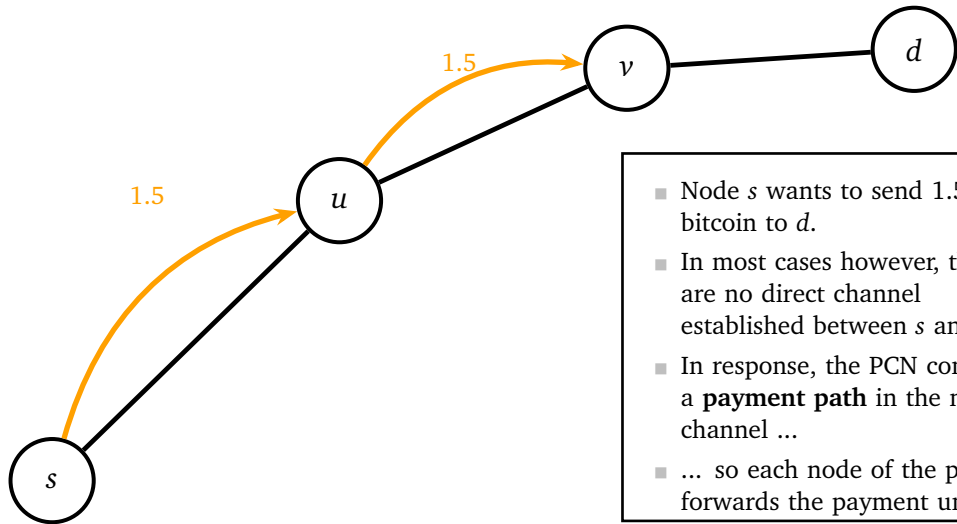
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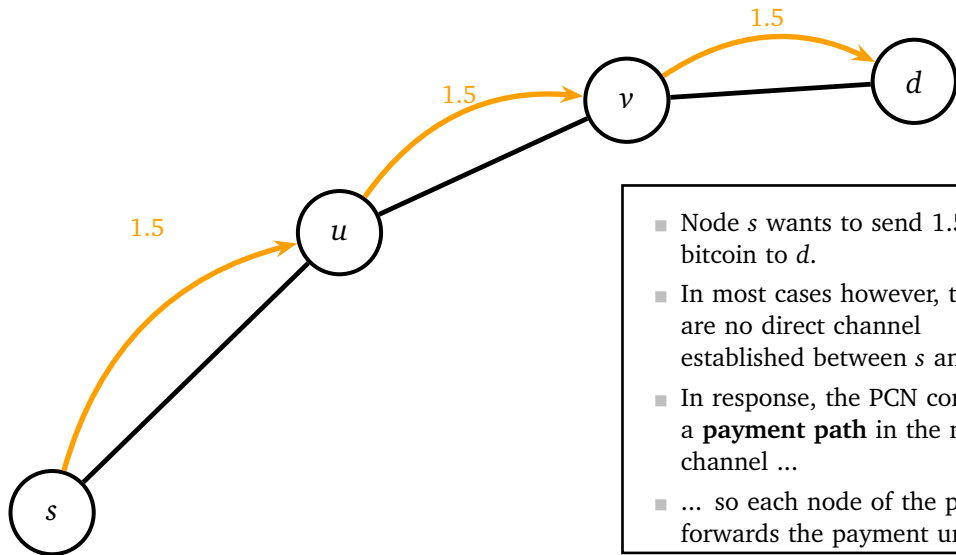
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Keeping the channels of a PCN balanced is considered the main bottleneck towards a better transaction throughput [Awathare + 2021; Mercan + 2021; Sivaraman + 2020; Chatterjee + 2025; Schmid + 2023].

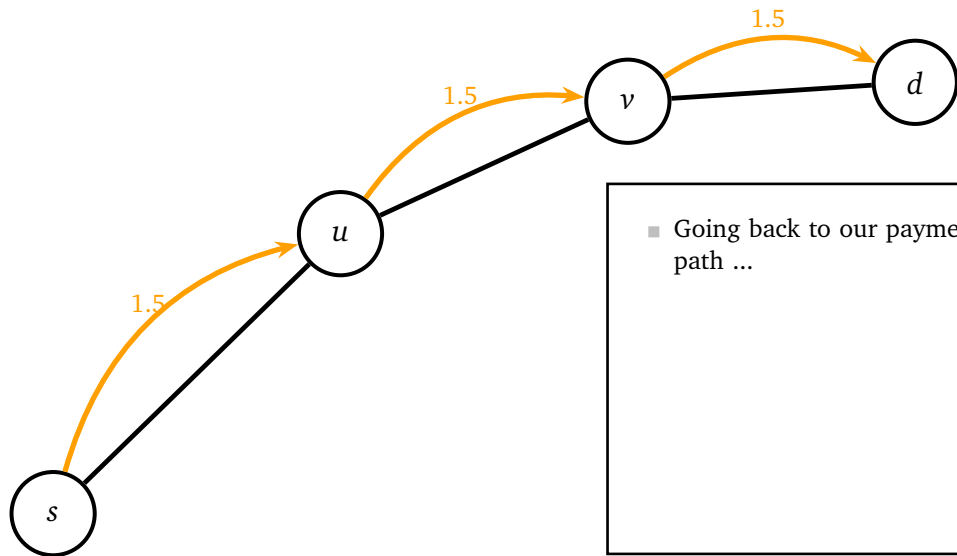
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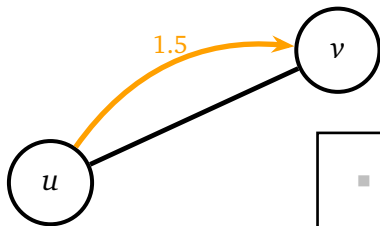
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⇒ The goal of our work is to **maximize the number of accepted transaction.**

The Transaction Admission Problem

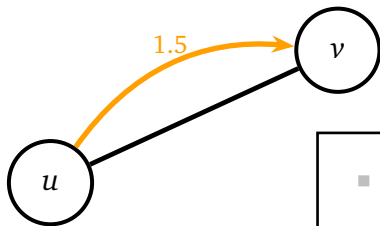


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The Transaction Admission Problem



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- ... let us look at the intermediary channel between u and v in isolation.
- Locally, payment proposals arrive from the outside. For each of those payment, **that channel decides whether to forward it or not.**

Algorithms for the Transaction Admission Problem

- **Greedy:** accepts any transaction that fits, used by virtually all (real time) algorithms in the literature. We show it has competitive ratio $\Omega(b)$, where v is the largest accepted transaction size.

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- **Greedy**: accepts any transaction that fits, used by virtually all (real time) algorithms in the literature. We show it has competitive ratio $\Omega(b)$, where v is the largest accepted transaction size.
- **Our algorithm (Exp)**: compares the transaction size with an (exponential) threshold that depends on the balance. We prove Exp is $O(\log b)$ -competitive. We prove this is **optimal** (even among randomized algorithms).

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From PCN to Knapsack

The transaction admission problem can be modelled as a **new variant of the online knapsack problem** where items can be **positive or negative**.

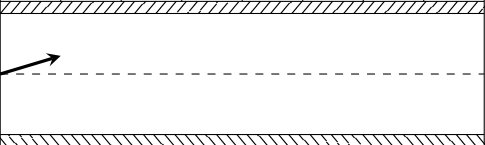
Here is a run of the greedy algorithm.

Transactions								
Accept ✓ or Reject ×								
Balance ($B = 10$)								

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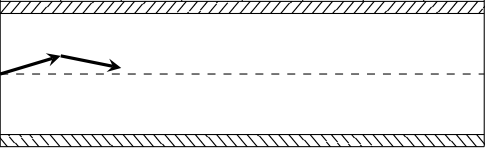
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Transactions	+3							
Accept ✓ or Reject ×	✓							
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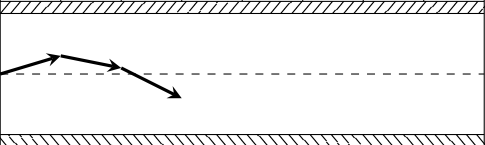
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Transactions	+3	-2						
Accept ✓ or Reject ×	✓	✓						
Balance ($B = 10$)	 The diagram shows a horizontal bar representing a balance. The bar is divided into three sections. The top section is shaded with diagonal lines. The middle section is white and contains a dashed horizontal line. The bottom section is shaded with diagonal lines. Two arrows point from the top of the bar to the dashed line, indicating a decrease in balance. The first arrow starts at the left edge of the bar and points to the dashed line. The second arrow starts at the dashed line and points to the right, indicating the current balance level.							

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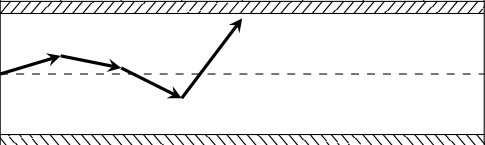
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Transactions	+3	-2	-5					
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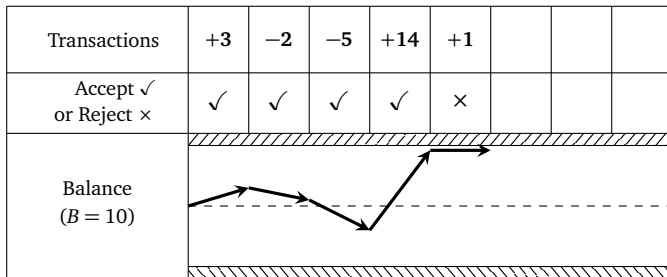
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Transactions	+3	-2	-5	+14				
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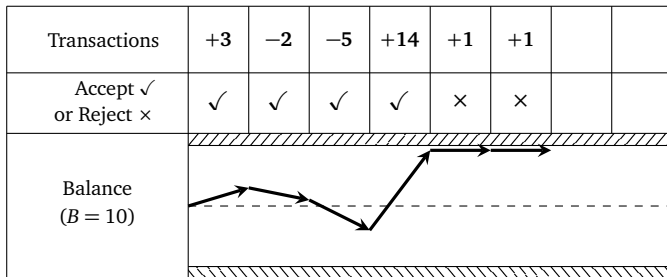
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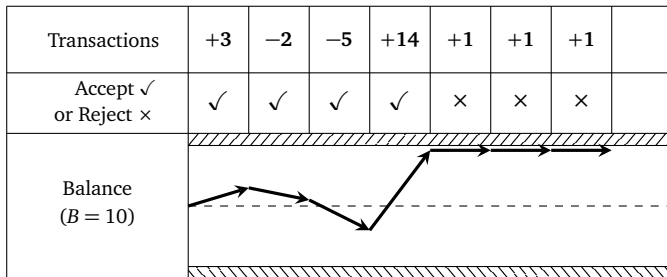
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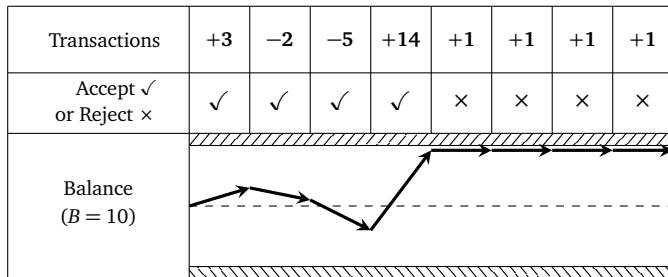
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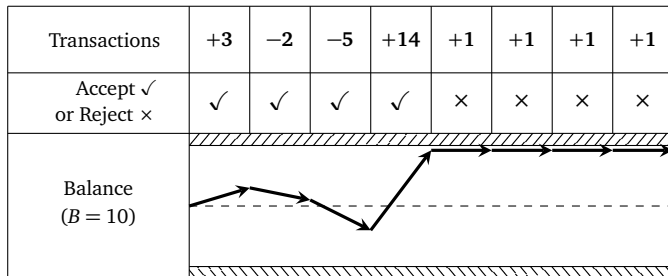
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⇒ we should not have accepted the fourth item.

Problem: Online Knapsack with Signed Items

Input

- Capacity $B \geq 1$, max item size $b \leq B$, initial state $s_0 \in [-B, B]$
- Items $\sigma_1, \sigma_2, \dots$ arrive online and one by one, each $\sigma_i \in [-b, -1] \cup [1, b]$

Possible Actions

Upon receiving σ_i with current state s , one can either:

- **Accept:** update $s \leftarrow s + \sigma_i$ (feasible only if $s + \sigma_i \in [-B, B]$)
- **Reject:** s stays unchanged

Objective & Measure

Maximize the number of accepted items. Algorithm Alg is **c-competitive** if, for some constant β ,

$$\forall \sigma, \quad c \cdot \text{Alg}(\sigma) \geq \text{Opt}(\sigma) - \beta$$

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- Our generalization to negative items is new. It makes the problem fundamentally harder as the input becomes inherently infinite.

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The Exp Algorithm

Insights on the algorithm

- Exp is **threshold-based**: it decides to accept if the incoming item is lower than a certain threshold
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Acceptance curve

Define $f(s) = b \cdot e^{-|s|/b}$. Accept σ_i iff:

1. σ_i and s have **opposite signs** (re-centering), **or**
2. $|\sigma_i| \leq f(s)$ (small enough)

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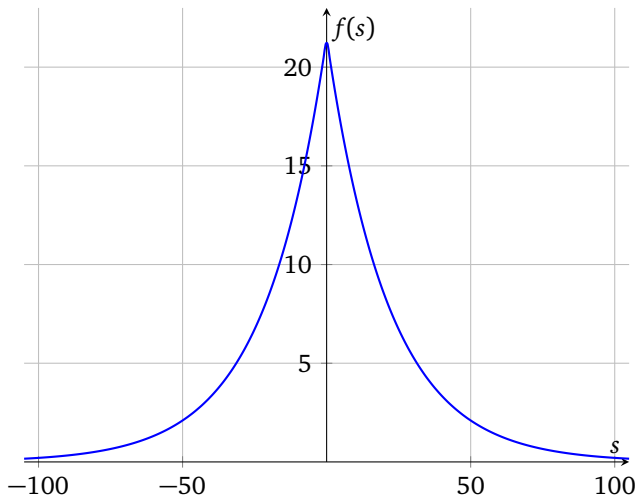
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Plot of function f for $B = 100$.

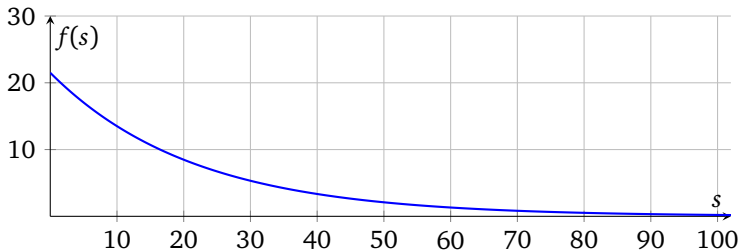
Intuition for the Exponential Curve

Key Insight

The exponential function is **invariant by truncation and then rescaling**:

$$\forall l > 0, \quad e^{-x} \Big|_{[0, +\infty]} \propto e^{-(x-l)} \Big|_{[l, +\infty]}$$

$\Rightarrow$ keeping only the part of the curve on the left of l gives the same shape. This fits our problem where the usable capacity shrinks as the state approaches B .



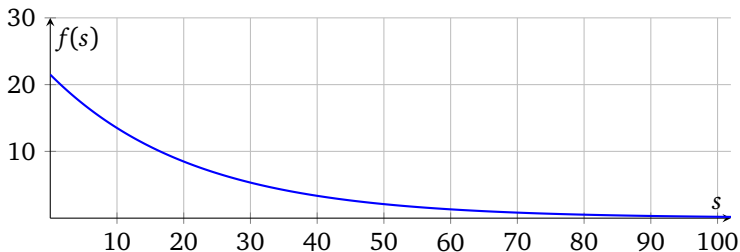
Right half of $f(s) = b e^{-s/b}$ for $B = 100$

Proof Idea: How Many Squares Can We Pack?

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Against our Exp algorithm, consider the following adversary: starting from $s = 0$, always request the largest item that is still accepted.

Visually, **this translates into a “square packing” problem.**



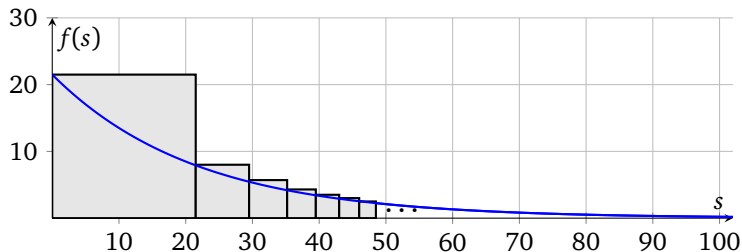
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Proof Idea: Main Steps

- Prove that the adversary packs at least $O(B/\ln B)$ squares under the curve.
- Prove that this square-packing sequence is indeed the worst-case input for Exp (with positive items only).
- Prove that mixing negative and positive items does not help the adversary — we handle this via a **potential function** argument.

Finally, we prove that **any algorithm**, even randomized, has a competitive ratio $\Omega(\log B)$, proving the asymptotic optimality of Exp.

Conclusion

What we did

- Modelled PCN channel admission as **online knapsack with signed items**
- Showed Greedy is $\Omega(b)$ -competitive (bad!)
- Proposed algorithm Exp: exponential acceptance threshold
- Proved Exp is $O(\log B)$ -competitive
- Proved $\Omega(\log B)$ lower bound for all randomized algorithms

Take-aways

- First **provably optimal** online algorithm for PCN transaction admission
- Simple to implement: $O(1)$ per transaction
- Works without knowing the input in advance
- Practical: competitive with Greedy on real data; better on adversarial inputs

Thank you for your attention!

References I

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